

Implications

$\Rightarrow$ means "implies"

e.g., $p \Rightarrow q$ means 'p implies q' or 'if p, then q'

$\Leftarrow$ means "implied by"

e.g., $p \Leftarrow q$ means 'p implied by q' or 'p given q'

$\Leftrightarrow$ means 'if and only if'

e.g., $p \Leftrightarrow q$ means 'p if and only if q' or 'p iff q'

P	Q	$P \Rightarrow Q$
F	F	T
F	T	T
T	F	F
T	T	T

P	Q	$P \Leftarrow Q$
F	F	T
F	T	F
T	F	T
T	T	T

P	Q	$P \Leftrightarrow Q$
F	F	T
F	T	F
T	F	F
T	T	T